

S M MASRUK UDDIN

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EDUCATION

Paris Sciences & Lettres (PSL) University

Master's in Space Science and Technology (International Research Track)

Institute: Observatoire de Paris

September 2023 – June 2025

Independent University, Bangladesh

January 2018 – April 2022

Bachelor of Science in Physics (Hons)

Department of Physical Sciences (Physics Major Program)

Minor in Computer Science

PUBLICATIONS

- Boin, T., Casamiquela, L., Haywood, M., Di Matteo, P., Lebreton, Y., Reese, D. R., Uddin, M., *Stellar Age Determination Using Deep Neural Networks- Stellar age determination using deep neural networks Isochrone ages for 1.3 million stars, based on BaSTI, MIST, PARSEC, Dartmouth, and SYCLIST evolutionary grids. Astronomy & Astrophysics* Accepted in March 2026. [arXiv:2603.09540](#)

Conference Proceedings

- Uddin, M., Boin, T., Casamiquela, L., Reese, D. R., Haywood, M. (2025), *Revisiting the Age Scale of Galactic Globular Clusters*. In *SF2A-2025: Proceedings of the Annual meeting of the French Society of Astronomy and Astrophysics*, ed. A. Siebert, pp. 315–317. [Astrophysics Data System \(ADS\) Entry](#)

Manuscripts in Preparation

- Uddin, M., Jash, R., Boin, T., Casamiquela, L., Reese, D. R., Haywood, M., *Revisiting the Age Scale of Galactic Globular Clusters*. In preparation (target journal: *Astronomy & Astrophysics*). *Updated results presented at SF2A 2025*;

RESEARCH

Master 2 — Lab Insertion Unit and Internship

September 2024 – July 2025

GEPI/LIRA (Observatoire de Paris)

Supervisors: Misha Haywood, Laia Casamiquela, Daniel Reese

Title: Revisiting the Age Scale of Galactic Globular Clusters

- Analyzed astrometric, spectroscopic, and photometric datasets (Gaia, APOGEE, HST) to derive precise ages for 50 Galactic globular clusters (HUGS catalog).
- Applied Bayesian inference with SPInS (MCMC) for stellar parameter estimation; Independently validating the results with a neural-network for fast age estimates.
- Investigating the age–metallicity relation to separate in-situ vs. ex-situ clusters.

Master 1 — Internship

January 2024 – June 2024

LUTH, Observatoire de Paris — Supervisor: E.ourgoulhon

Title: Gravitational Waves from Eccentric Orbits around Sgr A*

- Modeled GW signals from compact objects in eccentric orbits using the Black Hole Perturbation Toolkit.

- Compared Schwarzschild vs. Kerr cases; explored spin, eccentricity, and semilatus rectum effects on gravitational waveforms.
- Assessed LISA detectability and parameter constraints for such sources.

Master 1 — Lab Insertion Unit

December 2023

LUTH, Observatoire de Paris — Supervisors: E.ourgoulhon, L. Bernard

Title: Integrability of Geodesic Orbits around Black-Hole Spacetimes

- Studied differential geometry/tensor calculus with SageManifolds for GR. Explored geodesics in different coordinate to study singularity and it's behavior.

Senior-Year Project in Physics

Autumn 2021

Independent University, Bangladesh — Supervisor: M. A. Momen

Title: Infrared Divergence in Quantum Electrodynamics (QED)

- Reviewed strong-field QED (Breit–Wheeler) and infrared divergences at one loop; compared with charged scalar production.

Other Project

October 2020

The Effect of Second-Order Spin in Periastron Shift in Kerr Spacetime

DOI: [10.13140/RG.2.2.35341.36326](https://doi.org/10.13140/RG.2.2.35341.36326)

- Computed periastron shift contributions (Newtonian & relativistic) and applied to the S2–Sgr A* system;

CONFERENCE

Semaine de l'Astrophysique Française (SF2A)

July 1–4, 2025

INP–ENSEEIH, Toulouse, France

Poster: [Revisiting the Age Scale of Galactic Globular Clusters](#)

Authors: M. Uddin, R. Jash, T. Boin, L. Casamiquela, D. R. Reese, M. Haywood

Abstract: https://journees.sf2a.eu/wp-content/uploads/inscription/Internship_Abstract_Masruk.pdf

ACADEMIC ACHIEVEMENTS

- **International Mobility Grant** — PSL Graduate Program in Astrophysics 2023–2024
- **Cum laude**, 24th IUB Convocation Nov 8, 2023
- **IUB Merit Scholarship** Sep 2018 – Dec 2021
- **Honours Lists:** Dean's/Dean's Honour (2018), Dean's/Dean's Merit (2019), Vice Chancellor's (2020), Dean's (2021)

SKILLS

Programming: C/C++, Python, Java

HPC & Computing: Parallel programming, SLURM/job scheduling, large-scale Bayesian inference (~51,000 CPU hours on Paris Observatory's cluster: Tycho)

Software: Mathematica, TOPCAT, Autoplot, etc

Languages: English, Bengali, Chittagonian, Hindi, French (A1 in progress)

Documentation: L^AT_EX, Manim

SUMMER/WINTER SCHOOL

ICTP Physics Without Frontiers (Bangladesh)

Aug 25 – Oct 1, 2024

Teaching Assistant — School on Quantum Field Theory: <https://indico.ictp.it/event/10720>

India–Bangladesh Winter School in High-Energy Physics

Dec 17–21, 2019

EMPLOYMENT

Vice-Chancellor's Fellow & Course Contract Faculty

May 2022 – Jan 2023

Department of Mathematical and Natural Sciences, BRAC University, Dhaka

- Courses: PHY204 (Classical Mechanics & Special Relativity), MAT092 (Remedial Math), MAT110 (Differential Calculus & Coordinate Geometry), PHY111 (Mechanics, Properties of Matter), PHY112 (Electricity & Magnetism)

Student on Duty

Sep 2018 – Mar 2022

Physics Lab, Independent University, Bangladesh

OUTREACH ACTIVITIES

- **International Physics Olympiad (IPhO)** — Invigilator *2021*
- **Bangladesh Physics Olympiad** — Trainer & Academic Team *2017–2023*
- **Participation:** Bangladesh Physics Olympiad (2013–2016); Bangladesh Mathematical Olympiad (2012–2015)

REFERENCES

Dr. Misha Haywood

Astronomer, Observatoire de Paris – PSL (Meudon)
LIRA, Bureau 164, Grands Communs
5 Place Jules Janssen, 92195 Meudon, France
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Dr. Daniel Reese

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Dr. Laia Casamiquela

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Email: Laia.Casamiquela@obspm.fr